PROJECT OVERVIEW DOCUMENT

MineOps

Mining Operations Task Management & Safety Platform

Group: Syntax Crew

Module: I3691CP — Mobile Application Development

UNAM School of Engineering and the Built Environment | JEDS Campus

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1. What is MineOps?

MineOps is a mobile application built by Syntax Crew for the I3691CP semester project at UNAM's School of Engineering and the Built Environment. It is a real-time task management and safety platform designed specifically for mining operations.

The problem it solves:

Mining operations currently rely on paper logbooks, radio calls, and disconnected spreadsheets to communicate task status and hazard information between shifts. This creates Dead Zones — periods where incoming crews start work without knowing what was left unfinished, which hazards are active, or what equipment is out of service. MineOps replaces that broken system with a unified digital mobile platform where every task, hazard flag, and shift handover is recorded in real time and immediately visible to everyone who needs it.

2. Who Uses MineOps?

User Type	Who They Are	What They Do in MineOps
Site Supervisor	Manages daily operations across all zones	Creates tasks, assigns them to operators, monitors the live dashboard, generates shift handover reports
Operator / Crew	Field workers executing tasks on the ground	Views assigned tasks, updates task status, raises hazard flags when a safety issue is spotted
Compliance Officer	Ensures safety records meet regulatory standards	Exports hazard logs and audit reports for OSHA/MSHA compliance review
Fleet Manager	Tracks heavy machinery linked to tasks	Links equipment to tasks, receives maintenance alerts for machinery on active tasks
Administrator	IT/management — manages the system itself	Creates user accounts, assigns roles, configures site settings

3. How the App Works — Step by Step

3.1 Logging In

The user opens MineOps and sees a splash screen, then the Login screen. They enter their work email and password. Firebase Authentication checks the credentials. If correct, the app loads their role from Firestore (Supervisor, Operator, etc.) and takes them to the main dashboard. New users register via the Registration screen and are assigned a role by an Administrator.

3.2 The Dashboard — The Supervisor's Live View

The first thing a Supervisor sees after login is the live dashboard. It shows in real time:

- Total tasks for the current shift, broken down by status: Not Started, In Progress, Blocked, Completed.
- All active operators and the task each one is currently working on.
- All zones with active Hazard Flags, colour-coded by severity.
- Shift completion percentage — how much of today's planned work is done.
- Overdue tasks, highlighted in red.

The dashboard pulls data from Firestore and refreshes automatically every 5 seconds.

3.3 Tasks — The Core Workflow

A task is a unit of work assigned to an operator. Below is the full lifecycle of a task in MineOps:

Step	Who	What Happens
1	Supervisor	Opens Create Task. Fills in: title, zone, assigned operator(s), priority (Critical/High/Medium/Low), start and due date/time, and category (Drilling, Blasting, Haulage, Maintenance, Safety Inspection).
2	System	Saves the task to Firestore. Sends a push notification to the assigned operator via Firebase Cloud Messaging.
3	Operator	Opens My Tasks. Sees the new task at the top of their list. Taps it to view the full details and checklist.
4	Operator	Starts work. Taps In Progress on the task detail screen. Firestore updates instantly — the Supervisor's dashboard reflects the change.
5	Operator	Finishes the work. Taps Completed. A confirmation checklist appears and must be ticked before the task closes.
6	System	Marks the task Completed with a timestamp and acting user ID. Writes a permanent entry to the audit_logs collection.
7	System	If the task is past its due date and still open, the system escalates it — turns it red on the dashboard and notifies the Supervisor.

3.4 Hazard Flags — Safety Reporting

Any logged-in user can raise a Hazard Flag at any time. They open the Hazards tab, fill in the zone and description, set the severity level (Low, Medium, High, Critical), and optionally attach a photo (stored in Firebase Storage).

When a Critical flag is raised, every supervisor on site receives an immediate push notification. Tasks in the affected zone are automatically put On Hold. A supervisor must review and manually override the hold before work resumes. This prevents any worker from entering a hazardous zone due to a missed communication.

3.5 Shift Handover — Closing a Shift Safely

At the end of a shift, the outgoing Supervisor opens the Handover tab. MineOps automatically generates a Shift Handover Report from live Firestore data. The report includes:

- All tasks completed during the shift.
- All tasks still in progress or blocked, with documented reasons.
- All active Hazard Flags and their current resolution status.
- Equipment status notes.
- The Supervisor's own handover notes for the incoming shift.

The outgoing Supervisor digitally signs the report. The incoming Supervisor receives it as a push notification and must acknowledge receipt within 15 minutes. The report is stored permanently in Firestore and can be exported as a PDF for regulatory auditing.

3.6 Offline Mode

Mine environments often have poor connectivity. MineOps operates offline-first: operators can view tasks, update status, add notes, and raise hazard flags without a connection. All actions are queued locally on the device. When connectivity returns, the app automatically syncs all queued changes to Firebase. A persistent banner shows the user whether they are Online, Syncing, or Offline at all times.

4. Technology Stack

Technology	What It Is	How MineOps Uses It
Expo + React Native	JavaScript framework for mobile apps	Builds the entire app UI and logic. One codebase for Android and iOS.
Firebase Authentication	Google user login service	Handles registration, login, logout, and session management.
Cloud Firestore	Google real-time cloud database	Stores all tasks, hazard flags, handover reports, notifications, and audit logs.
Firebase Storage	Google cloud file storage	Stores photos attached to hazard reports and task updates.
Firebase Cloud Messaging	Push notification service	Delivers real-time alerts to operators and supervisors.
AsyncStorage	Local device storage	Queues actions locally when the device is offline.
EAS Build	Expo's cloud build service	Compiles the final Android APK for the Phase 4B submission.

GitHub	Version control platform	Hosts all code, documentation, and design files with full commit history.

5. Firebase Data Model

All app data lives in Firestore as collections. Below is every collection MineOps uses and what it stores:

Collection	What It Stores	Key Fields
users	One record per registered user	name, email, role, siteId, createdAt
tasks	Every task ever created	taskId, title, zone, assignedTo, priority, status, category, startDate, dueDate, createdBy, updatedAt
hazard_flags	Every hazard report raised	flagId, raisedBy, zone, severity, description, photoUrl, status, createdAt, resolvedAt
handover_reports	Shift handover reports	reportId, shiftDate, outgoingSupervisor, completedTasks, incompleteTasks, activeHazards, supervisorNotes, signedOff
notifications	Alerts sent to users	notifId, userId, type, message, isRead, createdAt
audit_logs	Immutable record of every system action	logId, actingUser, action, targetId, targetType, timestamp

6. Project Team

Role	Name	Student Number	Domain
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